

$$\begin{aligned}(A+B) \cdot (A+B)' &= (A+B)' \quad (\text{Postulate 4(b)}) \\ &= (A+C) \quad (\text{Postulate 5(b)}) \\ &= A' \quad (\text{Postulate 2(a)})\end{aligned}$$

$$A \cdot A' = 0 \quad (\text{Postulate 4(b)})$$

(b) Suggest a procedure for a general proof. Can we use the duality principle after proving (b) of problem 2-16?

**Solution:** Let  $F(a_1, a_2, \dots, a_n)$  be the given Boolean function. List down the  $2^n$  minterms in ascending order. Form  $2^{n-1}$  consecutive pairs of minterms and by using the postulates 4(b), 5(b) and 2(a), remove the literal  $a_n$ . Again form  $2^{n-2}$  consecutive pairs of minterms and by using the postulates 4(b), 5(b) and 2(a), remove the literal  $a_{n-1}$ . Repeat this process until the expression  $a_1 \cdot a_1'$  remains. By postulate 4(b),  $a_1 \cdot a_1' = 0$ .

The dual of  $(A'B'C' + A'BC' + A'BC + A'BC' + AB'C' + ABC' + ABC)$  is  $(A+B+C) \cdot (A+B+C) \cdot (A+B+C) \cdot (A+B+C) \cdot (A+B+C)$ .  
 $\therefore$  For  $n=3$ , the sum of minterms and the product of minterms are duals of each other.

We can similarly show that for general  $n$ , the sum of minterms and the product of minterms are duals of each other.

**2-20:** A majority gate is a digital circuit whose output is equal to 1 if the majority of the inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input majority gate. Simplify the function.

**Solution:**

| x | y | z | F |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}F &= x'y'z + xy'z + xy'z + xy'z \\ &= x'y'z + xy'z + xy'z + (xy'z + xy'z) \quad (\text{Theorem 1(a)}) \\ &= (x'y'z + xy'z + xy'z + xy'z) + xy'z \quad (\text{Theorem 4(a)}) \\ &= xy'z + (x'y'z + xy'z + xy'z + xy'z) \quad (\text{Postulate 3(a)})\end{aligned}$$